

How the Allocation of Lander Functions Impacts Human Lunar Exploration Architecture Propellant Demands

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As NASA looks to the moon as a proving ground for the technologies required to conduct the first human exploration campaign of Mars, this study assesses the potential propellant demands that a sustained human lunar presence would require. In particular, this study evaluates how the allocation and distribution of vehicle functions to a different number of vehicles can impact the propellant required to conduct a human lunar exploration mission. Four different transportation architectures for human lunar exploration and three different refueling assumptions were defined. The impact on the propellant required for a mission was assessed for all architectures and refueling assumptions for ranges of crew payload mass, vehicle inert mass fraction, and a sensitivity to the mission maneuver velocity magnitudes. The results of this analysis show that aggregating vehicle functions to a single-stage lander vehicle can have a positive impact on the architecture's level of vehicle reusability and that an increase in the number of refueling opportunities reduces the mass of the vehicles and of the propellant required per mission.

I. Nomenclature

AE	=	Ascent Element
CSM	=	Command and Service Module
ΔV	=	Velocity change magnitude for orbital maneuver
DE	=	Descent Element
HLS	=	Human Landing System
IMF	=	Inert Mass Fraction, a factor that influences a vehicle's performance/capability
I_{sp}	=	Specific Impulse, a measure of a vehicle's propulsive performance
ISRU	=	In-Situ Resource Utilization
LDAC	=	Lander Design Analysis Cycle
LEO	=	Low Earth Orbit
LLO	=	Low Lunar Orbit
LM	=	Lunar Module
LOI	=	Lunar Orbit Insertion
Mg	=	Mega-gram, a mass unit equivalent to metric tons
NRHO	=	Near-Rectilinear Halo Orbit
RTE	=	Reusable Transfer Element
SSL	=	Single-Stage Lander
TEI	=	Trans-Earth Injection

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II. Introduction

NASA is planning to return humans to the Moon in 2024, with the intent to sustain an ongoing presence on the lunar surface and to prepare for future crewed missions to Mars [1]. Currently, it is expected that the propellant needs for these missions will be met by delivery from Earth, either integrated as part of the vehicle or transferred from a tanker or depot. Production of the propellant at or near the location of use via in-situ resource utilization (ISRU) is a promising capability; however, at this time, ISRU has not been incorporated into NASA's architectures except for technology demonstration missions. Current plans for propulsion technologies for these missions may not be compatible with lunar derived propellants, as the Moon lacks abundant known quantities of carbon and nitrogen to support production of hydrocarbons or hydrazines. Further, the availability and accessibility of some resources, particularly lunar ice, are still mostly unknown. Finally, the exact propellant demand and demand locations will continue to change as NASA continues to develop the architectures, concepts of operations, and vehicle designs to support lunar exploration.

Previous studies have assessed the value to a campaign of how lunar ice, if available and accessible, could support human lunar missions [2]. The value of using lunar ice to produce oxygen and hydrogen propellant depends in part on the quantity of propellant needed per mission, the frequency of those missions per year, and the number of years the campaign continues, including whether that campaign transitions to supporting human Mars missions [3]. The frequency of missions and the number of years of the campaign depend on choices the United States makes about priorities in human space exploration, but the propellant needs of the mission depend in part on choices NASA makes around what systems to develop and how they will be used.

Thus, to inform future assessments of the value of lunar ISRU across a range of potential future architectures, there is a need to explore how architectural choices, particularly with respect to systems operating on or around the Moon, impact propellant demands. This paper will evaluate how the changes in the allocation of vehicle functions affect the propellant demands for a human lunar exploration architecture by evaluating the vehicles and the impact that vehicle sizing parameters and their payloads have on those demands. In doing so, it will answer the question: what is the range of propellant demands that could be expected to support the human exploration of Earth's moon? A subsequent paper will assess the trade in the cost-breakeven of supplying the propellant for the architectures via either delivery from Earth or production on the Moon.

Section III of this paper provides a background of lunar architectures and how vehicle functions were allocated in those architectures. Section IV defines the set of vehicle functions that are considered in this study. Section V describes the human lunar exploration architectures considered in this study that consist of one or more vehicles performing the previously listed functions. Section VI presents the assumptions and analysis method used to answer this paper's research question. Section VII will discuss the results of this analysis. Section VIII will summarize conclusions that can be drawn to inform future analyses and plans for human space exploration.

III. Background

A. The Apollo Program

The last of six successful NASA missions to transport humans to the lunar surface and back lifted off from Pad A of Launch Complex 39 at Kennedy Space Center on December 7th, 1972. The lunar segment of the Apollo missions was completed using the following vehicles: a Command and Service Module (CSM) and a Lunar Module (LM) consisting of a Descent Stage and an Ascent Stage. [4]

The CSM, after separation from the LM, remained in lunar orbit with the command module pilot. The purpose of the CSM was to sustain three crew members throughout the mission, enable crew transfer to and from the LM's ascent stage, and conduct lunar orbit insertion (LOI) and trans-earth injection (TEI) burns. [4]

The LM would descend to the surface piloted by the lunar module pilot and mission commander. The purpose of the LM's descent stage was to provide the necessary velocity change to de-orbit, descend, and land on the lunar surface, while transporting and protecting the ascent stage of the LM. The purpose of the LM's ascent stage was to provide life support to two crew members, act as a habitat throughout the duration of the lunar surface mission, and to transport the crew members back to lunar orbit and rendezvous with the CSM. The average amount of propellant required (with margin) for lunar descent and ascent were 8597 kg and 2377 kg, respectively to travel from an ≈ 110 km altitude quasi-circular orbit [4]. Both of the LM's stages combusted a mixture of storable propellants (dinitrogen tetroxide and Aerozine 50) [5]. No plans to enable vehicle refueling within a mission were implemented.

B. The Constellation Program

Between 2005 and 2009, the dream to return human crew to the lunar surface was briefly revived with the Constellation Program. The primary differences relative to an Apollo mission were the allocation of maneuvers to vehicles, an increase in the number of crew on a mission to four, and the addition of several new elements (like an airlock) [6, 7]. The primary elements required for the lunar segment of a Constellation program mission were the Altair lunar lander and the Orion crew capsule.

The Orion crew vehicle was analogous to the Apollo program's CSM. Orion was a multi-purpose vehicle and its utility lay in its ability to sustain between two and six crew members throughout a range of potential missions in low Earth orbit (LEO) and in deep-space [7]. In the context of a Constellation-type lunar surface mission, Orion and its service module were primarily used for life-support and communications up until the trans-earth injection burn [8]. The propulsion system in the service module used storable propellants [5].

The Altair lunar lander was analogous to the Apollo program's Lunar Module; it was made up of a descent and an ascent stage that are designed to transport a crew of four to and from the lunar surface for a seven day surface mission [7]. The ascent stage was used by the crew for their life-support and habitation throughout the mission, and to transport the crew members back to lunar orbit to rendezvous with the Orion crew vehicle. The ascent element used storable propellants for its propulsion system [7]. The descent element, on the other hand, used a higher efficiency liquid oxygen and liquid hydrogen propulsion system and was therefore used for the LOI burn as well as powered descent [6, 7]. The LDAC-2 (Lander Design Analysis Cycle-2) Safety Enhanced Altair lunar lander concept was designed to store 24859 kg and 2782 kg of propellant in its descent and ascent stages respectively (for the outpost design reference mission) [5]. No assumption to enable vehicle refueling within a mission was implemented.

C. The Artemis Program

The latest human lunar return program was announced in 2017 and will return humans to the Moon in 2024, with the intent to sustain an ongoing presence on the lunar surface and to prepare for future missions to Mars [1]. NASA has awarded base period contracts to three companies to develop a Human Landing System (HLS) after having completed a preliminary assessment [9]. For the ground-rules and assumptions that were defined, NASA developed a government reference design that includes three lunar lander stages: an ascent module, a descent module, and a transfer vehicle [10]. No details or mass characteristics have yet been published for the government reference HLS ascent module, descent module, or transfer vehicle.

The ascent module is analogous to the Apollo LM ascent stage and the Altair ascent stage. Its main purpose is command and communications, life-support, and habitation for the crew throughout the duration of the descent, surface, and ascent segments of the crewed lunar mission. The descent module is analogous to the Apollo LM descent stage and the Altair descent stage. The main purpose of the HLS descent module is to provide powered descent to the surface.

The transfer vehicle is intended to become a fully reusable vehicle that would provide part of the in-space maneuvering and descent from a cislunar staging orbit towards the lunar surface; this reduces the amount of propellant that the descent module requires for descent, which in turn allows the descent element mass to be reduced [10]. Once the transfer vehicle has completed its mission, it returns to the cislunar staging orbit. A fully reusable transfer vehicle implies that in-space propellant transfer and propellant delivery to the cislunar staging orbit will be required.

IV. Vehicle Functions

In this study, the authors define a vehicle function as the vehicle's role within an architecture, with respect to two categories: mission segment and payload type. This study is primarily focused on the mission segment vehicle functions, and a follow on paper will address variations in the payload type vehicle functions; both vehicle function types are presented here for completeness.

The mission segment vehicle functions define the subset of trajectories and maneuvers that a vehicle (with its payload) must perform: **ascent**, for vehicles that transport payloads from the surface of a celestial body; **descent**, for vehicles that transport payloads to the surface of a celestial body; and **in-space**, for vehicles that transport payloads from one in-space node to another in-space node.

The payload type vehicle function describes the payload that a vehicle is designed to transport: **crew**, for vehicles that transport astronauts within human exploration architectures; **cargo**, for vehicles that transport assets within human and robotic exploration architectures; and **propellant**, for vehicles that transport propellant in support of human and robotic exploration architectures. Figure 1 shows a Venn diagram for each category of vehicle function, with overlapping areas representing the potential for vehicles that perform multiple functions.

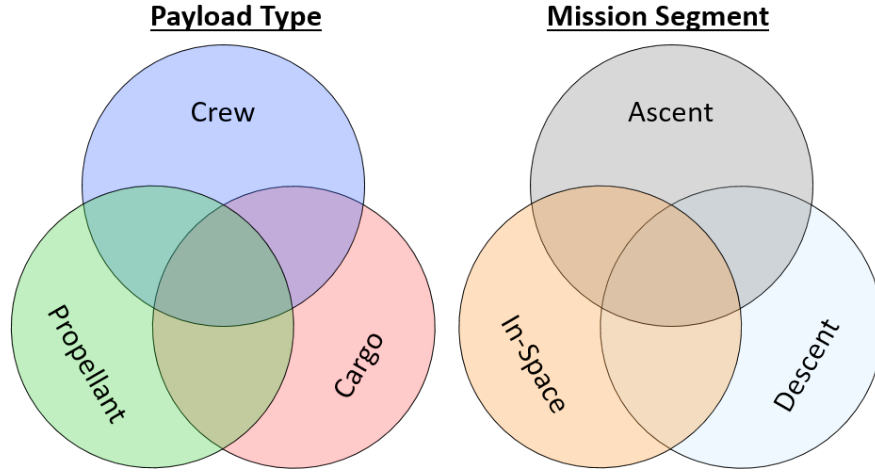


Fig. 1 Venn diagrams showing the combinations of vehicle function by their respective category: payload type and mission segment.

In an architecture, each vehicle must have at least one payload type function and at least one mission segment function. Vehicle functions have significant implications on the design and performance of the vehicles themselves and their integration within an architecture. While a vehicle designed and optimized for a crew payload could technically also transport cargo or propellant payloads, it is assumed that a vehicle designed specifically for either the cargo or propellant payload type would perform better than a crew payload vehicle transporting cargo or propellant or both; there are certain safety, habitation, and life support systems that are required to transport crew. From an architectural standpoint, it may be advantageous to have fewer vehicles that carry out more functions less efficiently than more vehicles that carry out fewer functions more efficiently.

This study examines the allocation of the mission segment functions across one or more vehicles for transporting crew, as the authors assume that crew transport will be the driving payload type in the formulation of human lunar exploration architectures. The follow-on paper will look at how the inclusion of the cargo and crew payload types can impact the assessment of lunar architectures, particularly by considering the possibility that having separate vehicles for cargo or propellant transport may be better for certain objectives than a single vehicle design for all three payload types.

V. Human Lunar Exploration Architectures

A set of four different human lunar exploration architectures were examined in this study, each having a different allocation and distribution of vehicle functions among the vehicles within the architecture. The architectures that were examined are a single-stage lander architecture, a single-stage lander with transfer element architecture, an ascent and descent element architecture, and a three-stage architecture (that includes an ascent, a descent, and a transfer element).

A. Single-Stage Lander Architecture

At the beginning of the mission, crew transfer from the cislunar aggregation node to the single-stage lunar lander vehicle. The single-stage lunar lander departs from the cislunar aggregation node with a separation and descent orbit insertion maneuver. The single-stage lander (SSL) conducts a powered descent maneuver to land on the lunar surface, where the human lunar exploration mission will take place. Upon completion of the surface mission, the SSL launches the crew from the surface and returns to and docks with the cislunar aggregation node. A diagram of this architecture is shown in Figure 2.

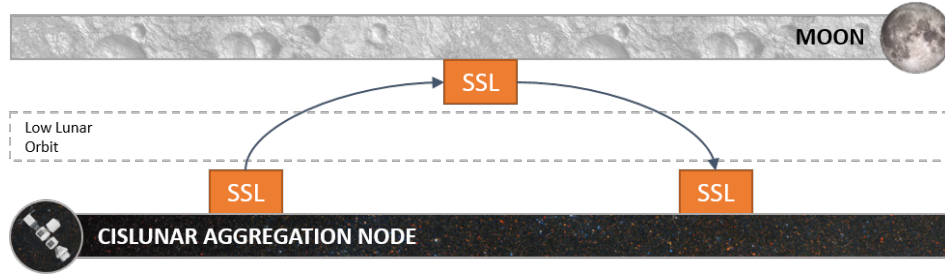


Fig. 2 A diagram of the single-stage lander human lunar architecture.

The SSL is therefore designed to transport a payload of crew (and crew-related cargo) from the cislunar aggregation node to the lunar surface and back; the allocation of vehicle functions is shown in Table 1. The single-stage lander is responsible for the following three vehicle functions: crew descent, crew ascent, and in-space transfer of crew from one node to another.

Table 1 Vehicle function allocation and distribution among the vehicles in Architecture A.

Vehicle	Vehicle Function		
	Crew – Ascent	Crew – Descent	Crew – In-Space
Single-Stage Lander	X	X	X

B. Single-Stage Lander With Transfer Element Architecture

At the beginning of the mission, crew transfer from the cislunar aggregation node to the SSL vehicle. The docked reusable transfer element (RTE) and SSL depart from the cislunar aggregation node with a separation and descent orbit insertion maneuver performed by the RTE. Once the two docked vehicles reach low lunar orbit (LLO) they separate; the RTE conducts a maneuver to return to the cislunar aggregation node for future reuse, while the SSL continues its descent by conducting a powered descent maneuver to land on the lunar surface. Upon completion of the surface mission, the SSL launches the crew from the surface and returns to and docks with the cislunar aggregation node. A diagram of this architecture is shown in Figure 3.

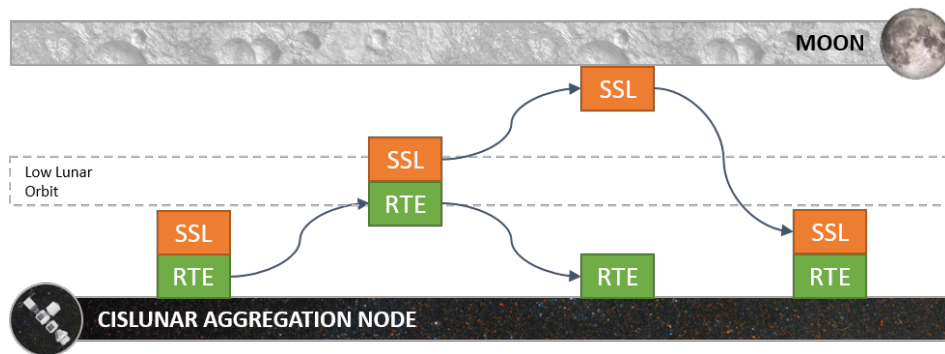


Fig. 3 A diagram of the lunar lander and transfer element human lunar architecture.

With the introduction of the RTE relative to Architecture A, the three vehicle functions are distributed among the two vehicles within the architecture, as shown in Table 2. The SSL is responsible for crew descent and crew ascent, whereas the RTE is responsible for the in-space transfer of crew.

Table 2 Vehicle function allocation and distribution among the vehicles in Architecture B.

Vehicle	Vehicle Function		
	Crew – Ascent	Crew – Descent	Crew – In-Space
Single-Stage Lander	X	X	-
Reusable Transfer Element	-	-	X

C. Ascent and Descent Element Architecture

At the beginning of the mission, crew transfer from the cislunar aggregation node to the ascent element (AE). The docked AE and descent element (DE) depart from the cislunar aggregation node with a separation and descent orbit insertion maneuver performed by the DE. The AE and DE remain docked through LLO and the entirety of the descent segment of the mission, as the DE conducts all powered descent maneuvers required to land on the lunar surface. Upon completion of the surface mission, the AE launches the crew from the surface and returns to and docks with the cislunar aggregation node. The DE, out of usable quantities of propellant, remains on the lunar surface. A diagram of this architecture is shown in Figure 4.

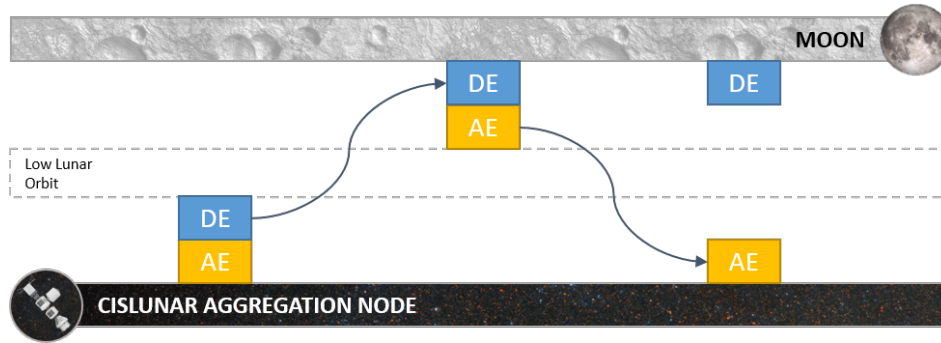


Fig. 4 A diagram of the ascent and descent element human lunar architecture.

The distribution of the three vehicle functions among the AE and DE is shown in Table 3. The AE is responsible for crew ascent and in-space transfer of crew, whereas the DE is responsible for crew descent and in-space transfer of crew.

Table 3 Vehicle function allocation and distribution among the vehicles in Architecture C.

Vehicle	Vehicle Function		
	Crew – Ascent	Crew – Descent	Crew – In-Space
Ascent Element	X	-	X
Descent Element	-	X	X

D. Three-Stage Architecture

At the beginning of the mission, crew transfer from the cislunar aggregation node to the AE. The docked AE, DE, and RTE depart from the cislunar aggregation node with a separation and descent orbit insertion maneuver performed by the RTE. The AE and DE remain docked through LLO, while the RTE undocks and conducts a powered maneuver to return to the cislunar aggregation node for future reuse. The DE conducts all powered descent maneuvers required for itself and the AE to land on the lunar surface. Upon completion of the surface mission, the AE launches the crew from the surface and returns to and docks with the cislunar aggregation node. The DE, out of usable quantities of propellant, remains on the lunar surface. A diagram of this architecture is shown in Figure 5.

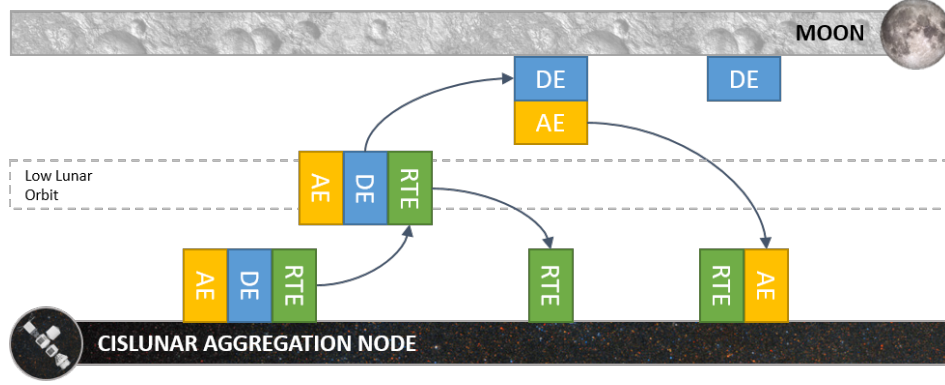


Fig. 5 A diagram of the ascent, descent, and reusable transfer element human lunar architecture.

The addition of the RTE to an AE/DE architecture distributes the vehicle functions as shown in Table 4. The AE is responsible for crew ascent, the DE is responsible for crew descent, and the RTE is responsible for in-space transfer of crew.

Table 4 Vehicle function allocation and distribution among the vehicles in Architecture D.

Vehicle	Vehicle Function		
	Crew – Ascent	Crew – Descent	Crew – In-Space
Ascent Element	X	-	X
Descent Element	-	X	-
Reusable Transfer Element	-	-	X

VI. Assumptions and Sizing Methods

A. Transportation Architecture Maneuvers

Each of the human lunar exploration architectures offers a different way of transporting crew (and their mission-related cargo) from an aggregation node in cislunar space to the lunar surface and back again via low-lunar orbit (LLO). The cislunar aggregation node is assumed to be in a Near-Rectilinear Halo Orbit (NRHO). The magnitudes of the maneuvers that the crewed vehicles must make within the transportation architecture are shown in Table 5 [11]. While there are trade studies that could be analyzed with respect to the vehicle sizing relative to varying descent and ascent concepts of operation, this study focuses on the vehicle sizing based on a generalized ΔV budget. Further effects of the ΔV budget are covered in the Section VII.C.

Table 5 Velocity change (ΔV) magnitude required for the crewed spacecraft to travel between Near-Rectilinear Halo Orbit (NRHO), Low Lunar Orbit (LLO), and the lunar surface. [11]

From	To	ΔV (m/s)
NRHO	LLO	750
LLO	Surface	2000
Surface	LLO	2000
LLO	NRHO	750

B. Vehicles and Vehicle Sizing

Each of the four vehicles (SSL, AE, DE, and RTE) represent a different grouping of vehicle functions; some vehicles are more independently capable, while others rely on each other. The allocation and distribution of vehicle functions among the four listed vehicles were shown in Tables 1 to 4 from Sections V.A to V.D respectively.

Each vehicle is sized to minimize its inert mass and deliver its respective payload (whether that is crew, cargo, or another vehicle) through one or more maneuvers. Using the rocket equation, vehicles are sized based on the maneuvers they conduct, as well as their assumed payload mass, inert mass fraction (IMF), and Specific Impulse (I_{sp}). The propulsion systems of all vehicles combust liquid hydrogen and liquid oxygen and have an assumed specific impulse of 450 s. Other sizing parameters including the payload mass and the IMF will be shown in later sections.

C. Refueling Assumptions

Based on the transportation architecture variants shown for the human lunar exploration architecture in Figures 2 to 5, there are a range of refueling assumptions that can impact vehicle sizing. Refueling can occur on the lunar surface, in cislunar space, or in both locations. The set of refueling assumptions that will be studied in this paper are the following:

- Maximum cislunar refueling – all vehicles in all architectures are fully refueled in cislunar space.
- Partial cislunar and surface refueling – all vehicles in all architectures are fully fueled in the departure location of their next mission segment (i.e. the DE is fueled in space, the AE is fueled on the surface).
- Maximum surface refueling – all vehicles in all architectures that go to the surface are fully refueled on the lunar surface space. The vehicles that do not land on the lunar surface are refueled in cislunar space.

Table 6 shows how the refueling assumptions impact each of the vehicles in the defined human lunar exploration architecture variants.

Table 6 Refueling trade-space options for the human lunar exploration architecture variants showing the location where each vehicle receives the propellant it requires.

Refueling Assumption	Location	Architecture A	Architecture B	Architecture C	Architecture D
Maximum Cislunar Refueling	Cislunar	SSL	SSL & RTE	AE & DE	AE & DE & RTE
	Surface	-	-	-	-
Partial Cislunar and Surface Refueling	Cislunar	SSL	SSL & RTE	DE	DE & RTE
	Surface	SSL	SSL	AE	AE
Maximum Surface Refueling	Cislunar	-	RTE	DE	DE & RTE
	Surface	SSL	SSL	AE	AE

As shown in Table 6, the single-stage lander vehicle is listed as receiving propellant in both locations for the partial cislunar and surface refueling assumption; the lander receives propellant for descent in cislunar space and receives propellant for ascent on the lunar surface. This option is a byproduct of allocating both the crew descent and crew ascent vehicle functions to the same vehicle. If the crew descent and crew ascent vehicle functions are distributed to different vehicles, as in architectures C and D, the maximum surface refueling option is removed from the subsequent figures since the results are equivalent to the partial cislunar and surface refueling assumption case considering that the DE is not reusable. This study examines the steady state propellant demands for a human lunar exploration architecture, and does not account for the emplacement of the vehicles.

There are vehicle sizing implications to these refueling assumptions. Using the ascent and descent element architecture (C) as an example, if surface refueling is not an option, the ascent element will have to be refueled in cislunar space and delivered to the surface by the descent element fully fueled. If surface refueling is available, the descent element will not have to transport a fully fueled ascent element to the surface, leading to a reduction in descent element payload mass, allowing the descent element itself to be smaller. Similarly, the single-stage lander that is fully refueled either in cislunar space or on the lunar surface is larger than the single-stage lander that is refueled in both locations.

VII. Results and Discussion

A. Baseline Case

A baseline case was defined to evaluate how changes in the crew payload mass (from 2000 kg to 12000 kg) impact the surface and cislunar propellant demands for the range of defined architectures and refueling assumptions. The vehicle sizing inputs for the baseline case are shown in Table 7. The results derived from each combination of architecture and refueling assumption can be compared visually in Figure 6 and numerically in Table 8.

Table 7 Vehicle sizing input parameters used in the baseline case.

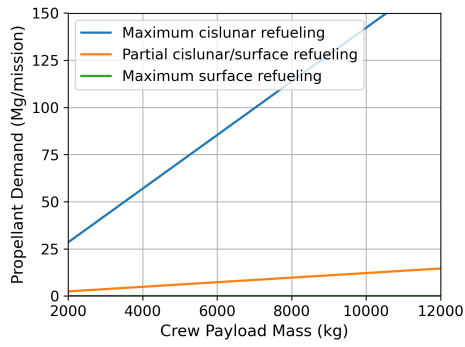
Parameter	SSL	AE	DE	RTE	Unit
Specific Impulse (I_{sp})	450.0	450.0	450.0	450.0	s
Inert Mass Fraction (IMF)	0.25	0.25	0.2	0.1	-

Table 8 Total propellant required per mission for each architecture and refueling assumption, where the minimum and maximum total propellant demand are associated with a crew payload mass of 2000 kg and 12000 kg respectively.

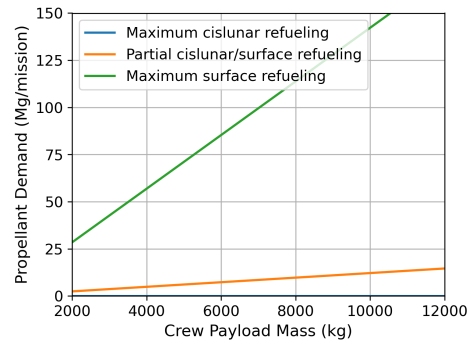
Architecture	Refueling Assumption	Total Propellant Demand		$\frac{\delta m_{prop}}{\delta m_{crew}}$
		Minimum (Mg)	Maximum (Mg)	
A	Maximum cislunar refueling	28.45	170.71	14.23
A	Partial cislunar and surface refueling	4.86	29.16	2.43
A	Maximum surface refueling	28.45	170.71	14.23
B	Maximum cislunar refueling	14.09	84.53	7.04
B	Partial cislunar and surface refueling	5.72	34.31	2.86
B	Maximum surface refueling	12.60	75.62	6.30
C	Maximum cislunar refueling	8.21	49.27	4.11
C	Partial cislunar and surface refueling	5.53	33.18	2.77
D	Maximum cislunar refueling	7.80	46.83	3.90
D	Partial cislunar and surface refueling	5.31	31.87	2.66

As shown in Figure 6, an increase in the crew payload mass yields an increase in the propellant required by and the overall wet mass of the vehicle or vehicles involved in the transportation of the crew payload. Increases in the masses of these vehicles can significantly limit the feasibility of an architecture when considering the deployment of those elements from Earth (not the focus of this paper).

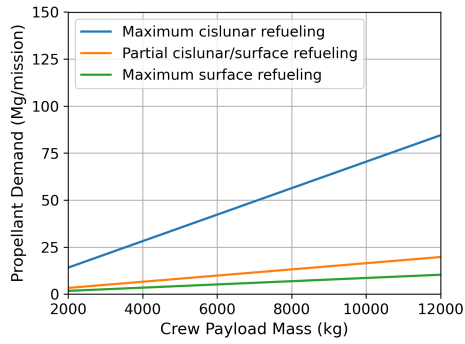
The differences in propellant demand between one architecture and another are noteworthy; however, a change in refueling assumptions can have a greater impact. For example in architecture A, switching from the maximum cislunar refueling or maximum surface refueling (which yield the same vehicle) to the partial cislunar and surface refueling assumption delivers significant vehicle mass and propellant demand savings: approximately six times less propellant is required for a mission (as shown in Table 8). For the other architectures, the same change in refueling assumptions would yield a reduction in the total propellant required for a mission; however, the magnitude of this change is smaller. Table 8 thus shows that distributing vehicle functions among multiple vehicles can yield reductions in the the propellant required for a mission. When the refueling opportunities are increased for the same set of architectures, aggregating vehicle functions to a single vehicle could be more optimal.



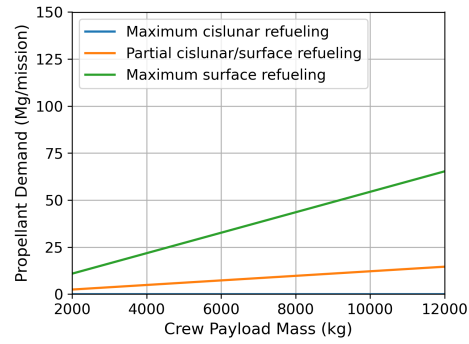
(a) Architecture A – Cislunar Propellant Demand



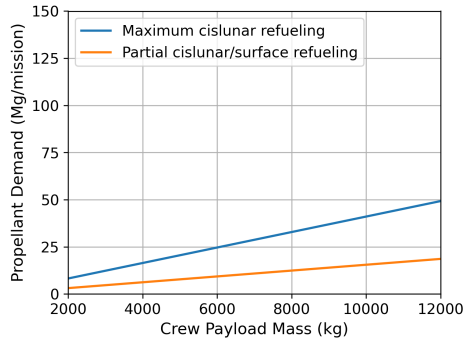
(b) Architecture A – Surface Propellant Demand



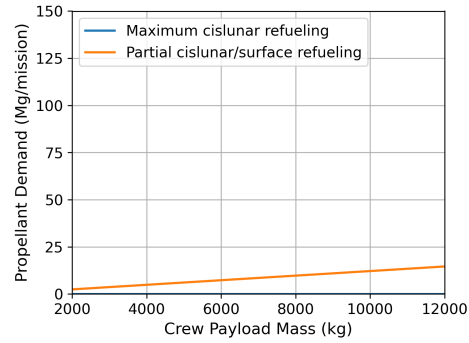
(c) Architecture B – Cislunar Propellant Demand



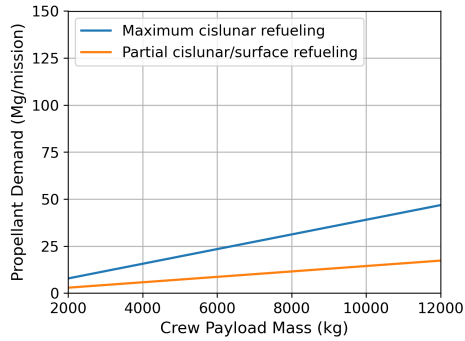
(d) Architecture B – Surface Propellant Demand



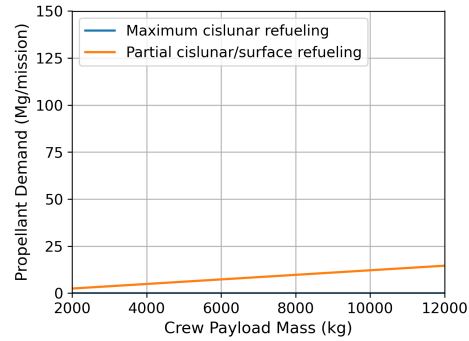
(e) Architecture C – Cislunar Propellant Demand



(f) Architecture C – Surface Propellant Demand



(g) Architecture D – Cislunar Propellant Demand



(h) Architecture D – Surface Propellant Demand

Fig. 6 The relationship between the crew payload mass and the surface/cislunar propellant demand for a range of architectures and refueling assumptions.

B. Varying the IMF

Fixing the crew payload mass to 4000 kg, a notional estimate for the Artemis Program, allowed evaluation of the impact of varying each of the vehicle inert mass fraction values independently. The IMF of the Single-Stage Lander and the Ascent Element were varied together. In each of the IMF sensitivities, shown in Sections VII.B.1 to VII.B.3, the case parameters shown in Table 9 were used to size the vehicles. While the IMF of a certain vehicle is not being varied, the mid-point of the stated range was used (equivalent to the IMF values shown in Table 7).

Table 9 Vehicle sizing input parameters used in the baseline case.

Parameter	SSL	AE	DE	RTE	Unit
Specific Impulse (I_{sp})	450.0	450.0	450.0	450.0	s
Inert Mass Fraction (IMF)	0.20–0.30	0.20–0.30	0.15–0.25	0.05–0.15	-

1. Single-Stage Lander and Ascent Element IMF

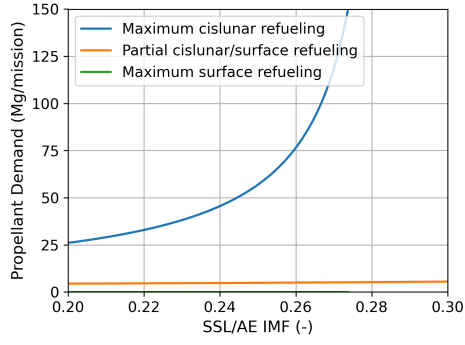
Varying only the single-stage lander and ascent element IMF from 0.20 to 0.30 yields the results shown in Table 10 and Figure 7.

Table 10 Total minimum and maximum propellant required per mission when varying the single-stage lander and ascent element IMF for each architecture and refueling assumption. The asterisk indicates the last data-point before the architecture no longer closes.

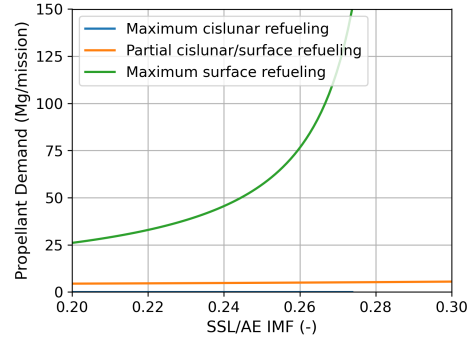
Architecture	Refueling Assumption	Total Propellant Demand	
		Minimum (Mg)	Maximum (Mg)
A	Maximum cislunar refueling	26.04	* 149.72
A	Partial cislunar and surface refueling	8.83	10.99
A	Maximum surface refueling	26.04	* 149.72
B	Maximum cislunar refueling	19.39	58.50
B	Partial cislunar and surface refueling	10.38	12.93
B	Maximum surface refueling	17.34	52.34
C	Maximum cislunar refueling	14.91	18.57
C	Partial cislunar and surface refueling	10.04	12.51
D	Maximum cislunar refueling	14.17	17.65
D	Partial cislunar and surface refueling	9.65	12.01

Varying the single-stage lander or ascent element IMF has a significant impact on all architectures and all refueling assumptions; increasing the IMF increases the total amount of propellant required to transport a 4000 kg payload to the lunar surface and back to the cislunar aggregation node. It should be noted that in the maximum cislunar refueling and the maximum surface refueling assumptions for Architecture A (see Figures 7a and 7b), increasing the single-stage lander and ascent element IMF beyond 0.275 yields transportation architectures that do not close; these are the only two cases where architecture closure cannot be achieved. As with the payload variation before, the transportation architecture that requires the least amount of propellant remains Architecture A with the partial cislunar and surface refueling assumption.

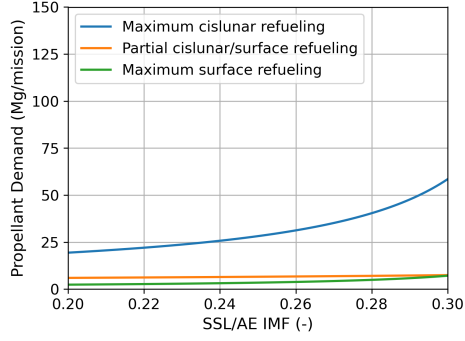
As can be seen in Figure 7 and Table 10, distributing vehicle function across an increased number of vehicles diminishes the impact that varying the single-stage lander and ascent element IMF has. The reason this particular IMF input is so much more influential than the others (as will be seen in Sections VII.B.2 and VII.B.3) is due to the order of operations: any change in the mass of a certain vehicle will influence any of the other vehicles that transport it, and subsequently any other vehicle even earlier in the transportation chain.



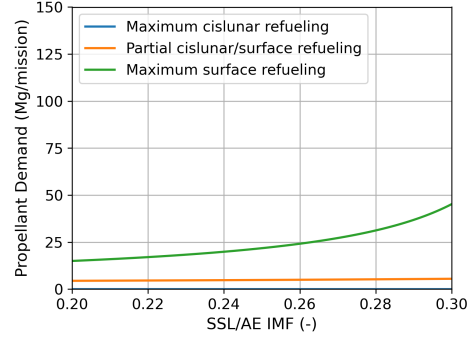
(a) Architecture A – Cislunar Propellant Demand



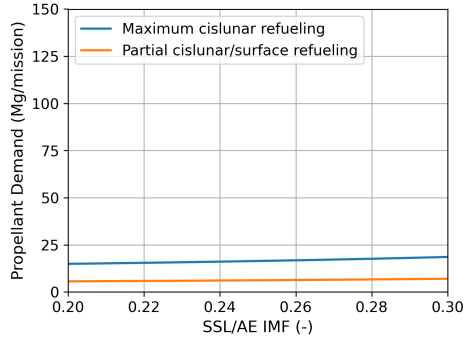
(b) Architecture A – Surface Propellant Demand



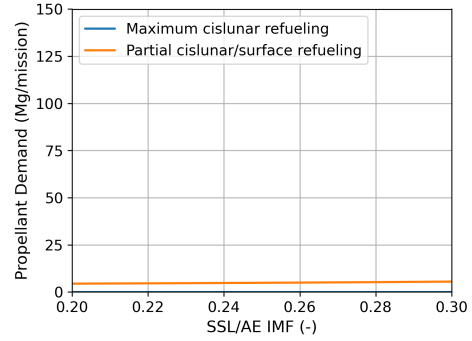
(c) Architecture B – Cislunar Propellant Demand



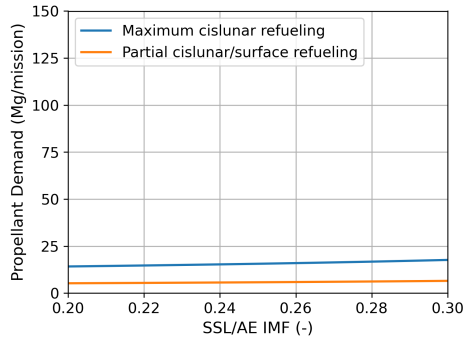
(d) Architecture B – Surface Propellant Demand



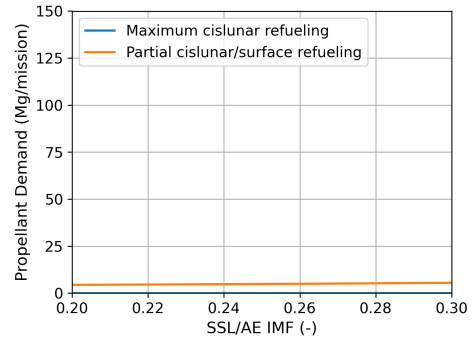
(e) Architecture C – Cislunar Propellant Demand



(f) Architecture C – Surface Propellant Demand



(g) Architecture D – Cislunar Propellant Demand



(h) Architecture D – Surface Propellant Demand

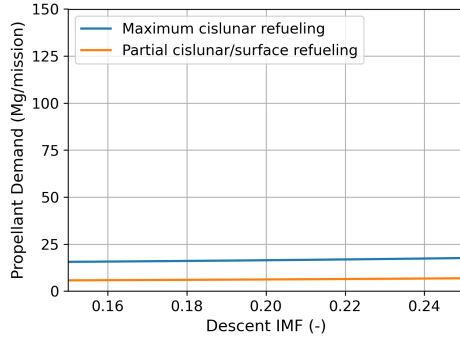
Fig. 7 The relationship between the single-stage lander and ascent element IMF and the surface/cislunar propellant demand for a range of architectures and refueling assumptions.

2. Descent Element IMF

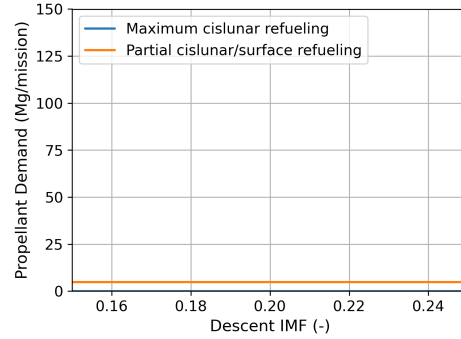
Varying only the descent element IMF from 0.15 to 0.25 yields the results shown in Table 11 and Figure 8. Because the descent element IMF only has an impact on architectures that use a descent element (Architectures C and D), all other architectures have been omitted from these results.

Table 11 Total minimum and maximum propellant required per mission when varying the descent element IMF for each applicable architecture and refueling assumption.

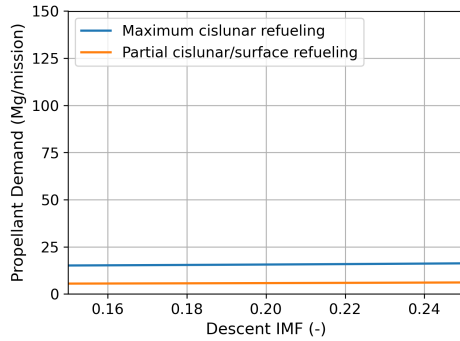
Architecture	Refueling Assumption	Total Propellant Demand	
		Minimum (Mg)	Maximum (Mg)
C	Maximum cislunar refueling	15.56	17.59
C	Partial cislunar and surface refueling	10.60	11.69
D	Maximum cislunar refueling	15.10	16.24
D	Partial cislunar and surface refueling	10.35	10.96



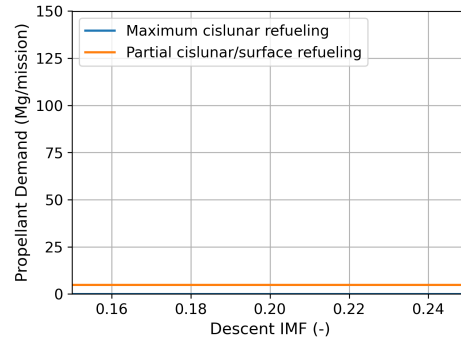
(a) Architecture C – Cislunar Propellant Demand



(b) Architecture C – Surface Propellant Demand



(c) Architecture D – Cislunar Propellant Demand



(d) Architecture D – Surface Propellant Demand

Fig. 8 The relationship between the descent element IMF and the surface/cislunar propellant demand for a range of architectures and refueling assumptions.

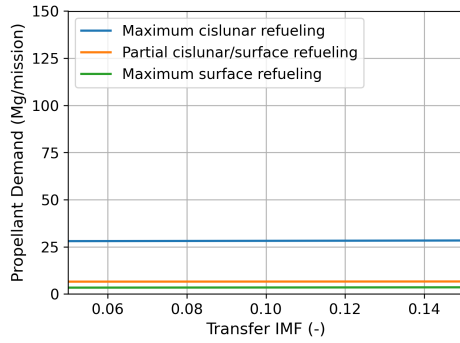
Varying the descent element IMF has a much less significant impact on the transportation architectures than varying the single-stage lander and ascent element IMF. This finding supports the previous determination that distributing vehicle functions over a larger number of vehicles can be beneficial since it mitigates the vehicle and propellant demand growth caused by increases in the descent element IMF. The descent element, when compared to the single-stage lander and ascent element, is a higher performance vehicle due to its lower IMF.

3. Transfer Element IMF

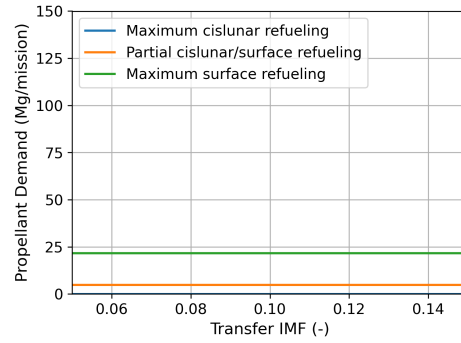
Varying only the transfer element IMF from 0.05 to 0.15 yields the results shown in Table 12 and Figure 9. Because the reusable transfer element IMF only has an impact on architectures that use a reusable transfer element (Architectures B and D), all other architectures have been omitted from these results.

Table 12 Total minimum and maximum propellant required per mission when varying the reusable transfer element IMF for each applicable architecture and refueling assumption.

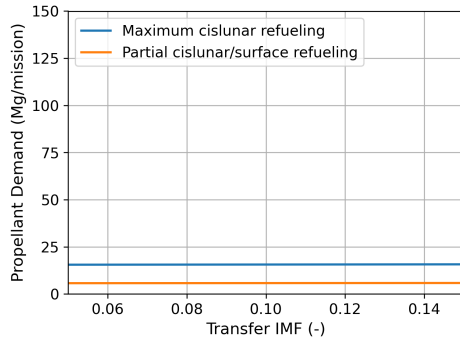
Architecture	Refueling Assumption	Total Propellant Demand	
		Minimum (Mg)	Maximum (Mg)
B	Maximum cislunar refueling	28.02	28.36
B	Partial cislunar and surface refueling	11.39	11.48
B	Maximum surface refueling	25.12	25.30
D	Maximum cislunar refueling	15.52	15.71
D	Partial cislunar and surface refueling	10.58	10.68



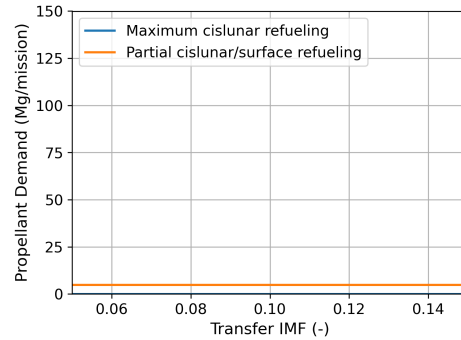
(a) Architecture B – Cislunar Propellant Demand



(b) Architecture B – Surface Propellant Demand



(c) Architecture D – Cislunar Propellant Demand



(d) Architecture D – Surface Propellant Demand

Fig. 9 The relationship between the transfer element IMF and the surface/cislunar propellant demand for a range of architectures and refueling assumptions.

Varying the reusable transfer element IMF has a negligible impact on the transportation architectures than varying the single-stage lander and ascent element IMF or the descent element IMF. The reusable transfer element, when compared to the single-stage lander and ascent element and descent element, is the highest performance vehicle in any of the architectures that include it.

C. Sensitivity to the ΔV Magnitudes

Assessing different combinations of ΔV variations from the baseline values shown in Table 5 shows the impact of that changes in the energy required to transport crew through an entire mission can have on the propellant demands. The ΔV variations studied in this paper are +5% and +10% from the baseline case, shown in Table 13, which can represent deviations from both nominal trajectories and nominal landing sites. The crew payload mass assumed for this case is 4000 kg and the vehicle sizing parameters used are shown in Table 7.

Table 13 Velocity change (ΔV) magnitude required for the crewed spacecraft to travel between NRHO, LLO, and the lunar surface. [11]

From	To	ΔV (m/s)		
		Baseline	+5%	+10%
NRHO	LLO	750.0	787.5	825.0
LLO	Surface	2000.0	2100.0	2200.0
Surface	LLO	2000.0	2100.0	2200.0
LLO	NRHO	750.0	787.5	825.0

Using the baseline case vehicle sizing parameters and a 4000 kg crew payload, the range of propellant demands for each architecture, refueling assumption, and ΔV case are shown in Table 14.

Table 14 Propellant required per mission for each combination of architecture, refueling assumption, and ΔV case.

Architecture	Refueling Assumption	Propellant Required (Mg)			Total Inert Mass (Mg)		
		Baseline	+5%	+10%	Baseline	+5%	+10%
A	Maximum cislunar refueling	56.90	108.45	-	18.97	36.15	-
A	Partial cislunar and surface refueling	9.72	10.68	11.73	1.62	1.78	1.95
A	Maximum surface refueling	56.90	108.45	-	18.97	36.15	-
B	Maximum cislunar refueling	18.28	21.19	24.65	3.83	4.39	5.05
B	Partial cislunar and surface refueling	14.29	15.79	17.45	2.13	2.35	2.59
B	Maximum surface refueling	13.56	15.42	17.58	3.31	3.75	4.26
C	Maximum cislunar refueling	16.42	18.70	21.30	4.51	5.12	5.81
C	Partial cislunar and surface refueling	11.06	12.28	13.64	3.17	3.52	3.90
D	Maximum cislunar refueling	15.83	17.97	20.41	3.12	3.50	3.93
D	Partial cislunar and surface refueling	10.74	11.90	13.19	2.43	2.67	2.95

As shown by the results in Table 14, variations in ΔV magnitudes can have significant effects on an architecture and refueling assumption. Most notably, for the maximum refueling assumptions (both cislunar or surface) of Architecture A, a 10% increase in ΔV magnitudes forces vehicle masses to increase to the point where the architectures no longer close (as marked by the dashes in Table 14). As vehicle functions are distributed over a larger number of vehicles, the impact that an increase in ΔV magnitudes has on the architecture is reduced. Additionally, an increase in the number of planned refueling locations throughout a mission also lessens the impact of an increase in ΔV magnitudes.

VIII. Conclusions

Certain allocations of vehicle functions and re-fueling options are more conducive to specific objectives. While this paper is not intended to determine whether one architecture is better than another, it does enable consideration of how these architectures differ in terms of their reliance on Earth, level of reusability, launch vehicle-related mass constraints, and potential cost effectiveness.

Architectures A and B would be the most suitable if the objectives were to enable reusability and have a reduced reliance on Earth. Architectures C and D require that a DE be launched from Earth for each mission, increasing their reliance on Earth and reducing the level of reusability that is achieved. On the other hand, if current launch vehicle constraints persist, the distribution of vehicle functions as shown in Architectures C and D would be enabling to accomplishing human lunar missions. However, this paper shows that distributing the vehicle functions over multiple vehicles is not the only viable solution for implementing a mass-constrained architecture; an alternate solution is to increase the number of refueling opportunities available for the vehicle(s) in the architectures. The inert mass fraction of the element performing ascent has a particular impact on the magnitude of these effects; thus, vehicle technology capabilities will also play an important role in choosing among different allocations.

A comparison of each architecture on the basis cost would require an evaluation of the human lunar exploration campaigns overall. Factors that would influence the cost include the number of launches that would be required, the frequency and number of vehicle replacements, and the duration of the campaign.

Therefore, future work should explore not just the propellant mass needs for human lunar missions, but also the costs of implementing those missions. Combined with an understanding of potential lunar campaigns, and thus the number and frequency of those missions, it would be possible to assess the costs of different architectural choices at a campaign level. Further, modeling of the source of that propellant for transfer (whether delivered from Earth or produced on the Moon) would allow for consideration of the value of implementing in-situ propellant production on the Moon as a function of choices in the allocation of vehicle functions. For these considerations, the modeling in this analysis can serve as a basis for understanding the impact of the allocation of vehicle functions on NASA's future plans to explore the Moon.

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References

- [1] Office of the Press Secretary, "National Space Policy of the United States of America," The White House, Washington D.C., 2017.
- [2] Jones, C. A., Klovstad, J., Judd, E., and Komar, D., "Cost Breakeven Analysis of Cis-lunar ISRU for Propellant," *AIAA Scitech 2019 Forum*, American Institute of Aeronautics and Astronautics, San Diego, California, 2019. doi:10.2514/6.2019-1372, URL <https://arc.aiaa.org/doi/10.2514/6.2019-1372>.
- [3] Jones, C. A., Pensado, A., Clark, M. A., Grande, M. L., Ivanco, M. L., Judd, E. L., Klovstad, J. J., and Reeves, D. M., "Cost Breakeven Analysis of Lunar In-Situ Propellant Production for Human Missions to the Moon and Mars," American Institute of Aeronautics and Astronautics, Las Vegas, 2020.
- [4] Orloff, R. W., "Apollo by the numbers: a statistical reference," National Aeronautics and Space Administration, Washington D.C., 2000.
- [5] Connolly, J. F., "NASA Lunar Lander Concepts Beyond Apollo," National Aeronautics and Space Administration, Washington D.C., 2009.
- [6] Kos, L., Polsgrove, T., Sostaric, R., Braden, E., Sullivan, J., and Le, T., "Altair Descent and Ascent Reference Trajectory Design and Initial Dispersion Analyses," *AIAA Guidance, Navigation, and Control Conference*, American Institute of Aeronautics and Astronautics, Toronto, Ontario, Canada, 2010. doi:10.2514/6.2010-7720, URL <http://arc.aiaa.org/doi/10.2514/6.2010-7720>.
- [7] Connolly, J. F., "Constellation Program Overview," National Aeronautics and Space Administration, Washington D.C., 2006.
- [8] Cohen, M., "From Apollo LM to Altair: Design, Environments, Infrastructure, Missions, and Operations," *AIAA SPACE 2009 Conference & Exposition*, American Institute of Aeronautics and Astronautics, Pasadena, California, 2009. doi:10.2514/6.2009-6404, URL <http://arc.aiaa.org/doi/10.2514/6.2009-6404>.
- [9] National Aeronautics and Space Administration, "NASA Selects Blue Origin, Dynetics, SpaceX for Artemis Human Landers," Apr. 2020. URL <https://www.nasa.gov/feature/nasa-selects-blue-origin-dynetics-spacex-for-artemis-human-landers>.

- [10] Smith, M., "Human Landing System BAA," National Aeronautics and Space Administration, Washington D.C., 2019. URL https://www.nasa.gov/sites/default/files/atoms/files/nasa_hls_baa_industry_forum_14feb2019.pdf.
- [11] Smith, M., "Advanced Exploration Systems - Deep Space Exploration," , Oct. 2019. URL <https://www.nasa.gov/sites/default/files/atoms/files/20191030-nac-heoc-smith-v2.pdf>.